Installation of Jenkins

What It Is

Jenkins is an **open-source automation server** mainly used for **CI/CD (Continuous Integration & Continuous Deployment)**.

If you're building projects (like your full-stack system), Jenkins automates:

- Building the code
- Running tests
- Creating Docker images
- Deploying to servers (EC2, Kubernetes, etc.)

We have to install java to run the **jetkins**.

```
ubuntu@ip-172-31-19-132:~$ sudo su
root@ip-172-31-19-132:/home/ubuntu# sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1446 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1741 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [324 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
```

```
i-00f9e6d4a01449e17 (jenkinservice)
PublicIPs: 98.94.2.97   PrivateIPs: 172.31.19.132
```

```
CloudShell   Feedback   Console Mobile App   © 2026, Amazon Web Services, Inc. or its affiliates.   Privacy   Terms   Cookie preferences
```

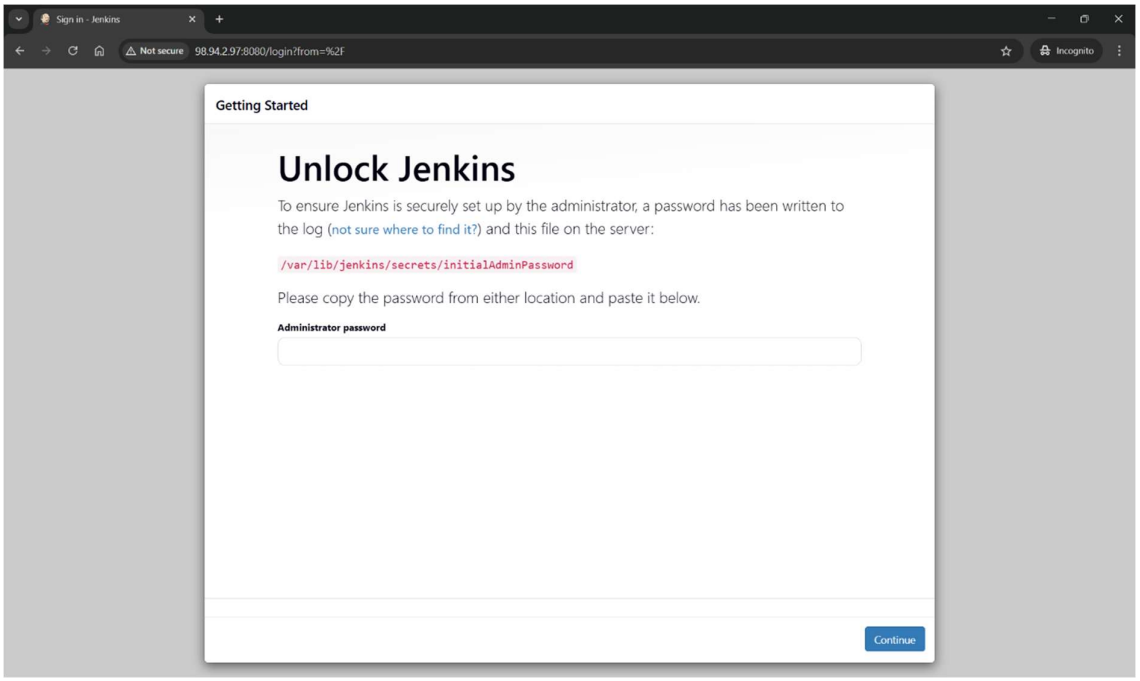
And then we have to install **Jenkins** and start the **Jenkins**.using the **systemctl start enable Jenkins** command.

```
                    tu# sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
  https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
--2026-02-11 12:47:03--  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.34.133, 2a04:4e42:78::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.34.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1.6K) [application/octet-stream]
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'

/etc/apt/keyrings/jenkins-keyring.asc         100%[===================================================================================>]   1.64K  --.-KB/s    in 0s

2026-02-11 12:47:03 (27.4 MB/s) - '/etc/apt/keyrings/jenkins-keyring.asc' saved [1680/1680]

Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:5 https://pkg.jenkins.io/debian-stable binary/ InRelease
```

```
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-19-132:/home/ubuntu# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-02-11 12:47:37 UTC; 19s ago
   Main PID: 4650 (java)
      Tasks: 42 (limit: 1008)
     Memory: 300.1M (peak: 321.7M)
        CPU: 17.924s
     CGroup: /system.slice/jenkins.service
             └─4650 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpport=8080

Feb 11 12:47:33 ip-172-31-19-132 jenkins[4650]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Feb 11 12:47:33 ip-172-31-19-132 jenkins[4650]: [LF]>
Feb 11 12:47:33 ip-172-31-19-132 jenkins[4650]: [LF]> *************************************************************
Feb 11 12:47:33 ip-172-31-19-132 jenkins[4650]: [LF]> *************************************************************
Feb 11 12:47:33 ip-172-31-19-132 jenkins[4650]: [LF]> *************************************************************
Feb 11 12:47:37 ip-172-31-19-132 jenkins[4650]: 2026-02-11 12:47:37.387+0000 [id=32]        INFO        jenkins.InitReactorRunner$1#onAttained: Completed initialization
Feb 11 12:47:37 ip-172-31-19-132 jenkins[4650]: 2026-02-11 12:47:37.412+0000 [id=24]        INFO        hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
Feb 11 12:47:37 ip-172-31-19-132 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Feb 11 12:47:37 ip-172-31-19-132 jenkins[4650]: 2026-02-11 12:47:37.491+0000 [id=50]        INFO        h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tools
Feb 11 12:47:37 ip-172-31-19-132 jenkins[4650]: 2026-02-11 12:47:37.494+0000 [id=50]        INFO        hudson.util.Retrier#start: Performed the action check updates server successfully

root@ip-172-31-19-132:/home/ubuntu# cat /var/lib/jenkins/secrets/initialAdminPassword
7b7cb76e96ca42af8c191a38402c12a8
root@ip-172-31-19-132:/home/ubuntu#
```

i-00f9e6d4a01449e17 (jenkinservice)
PublicIPs: 98.94.2.97 PrivateIPs: 172.31.19.132

Then when we open the public ip address we will see the Jenkins website for creation of **admin**.

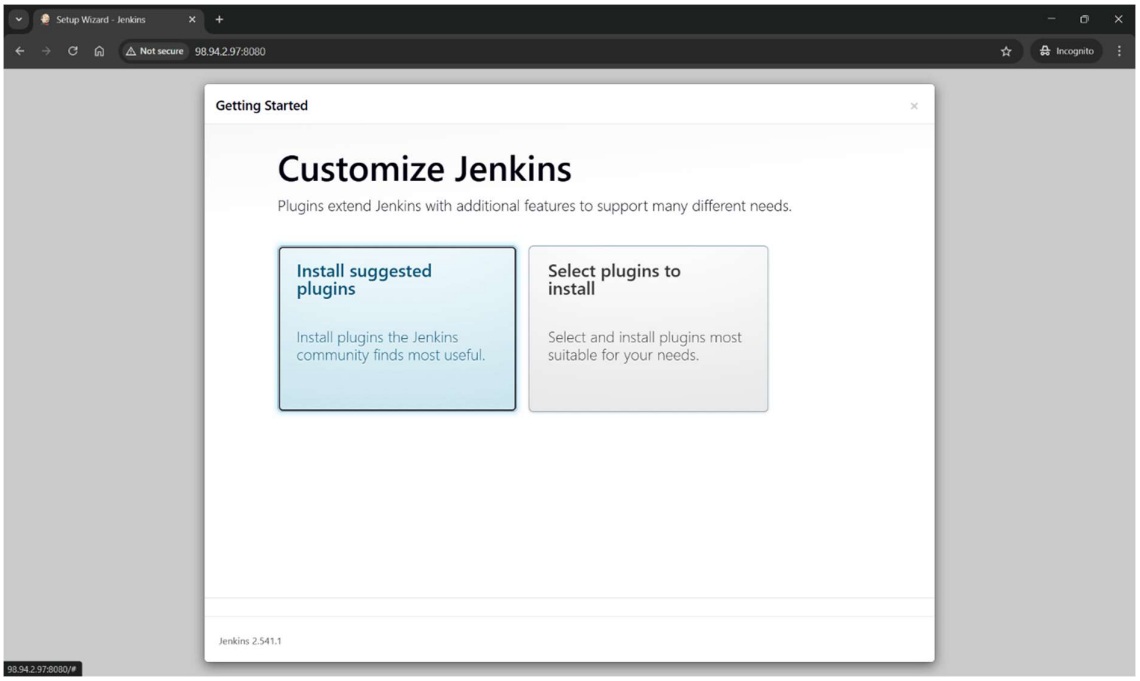


Then we have to get the password from the specified directory.

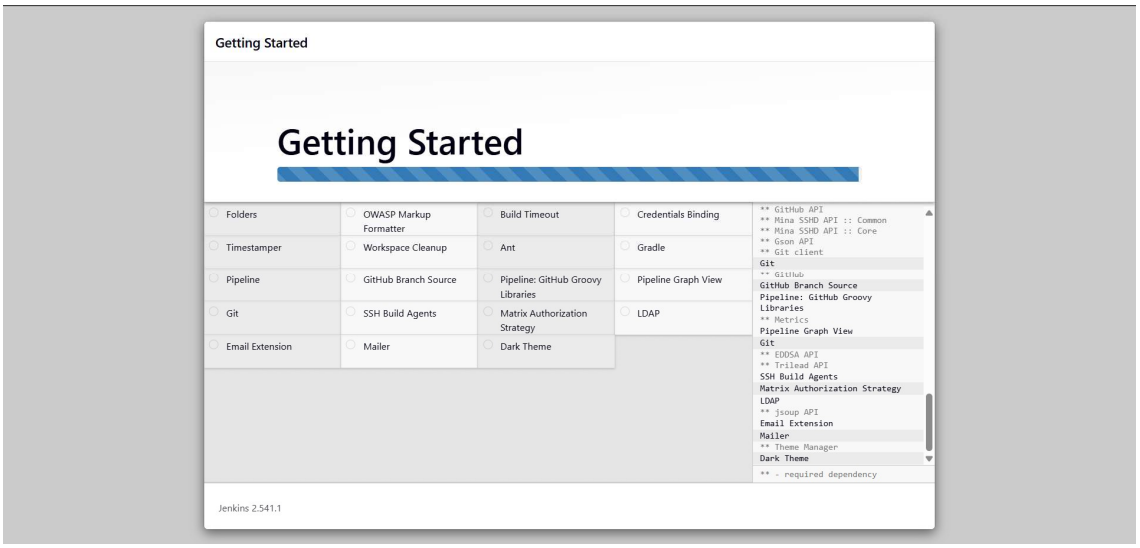
```
root@ip-172-31-19-132:/home/ubuntu# cat /var/lib/jenkins/secrets/initialAdminPassword
7b7cb76e96ca42af8c191a38402c12a8
root@ip-172-31-19-132:/home/ubuntu#
```

i-00f9e6d4a01449e17 (jenkinservice)
PublicIPs: 98.94.2.97 PrivateIPs: 172.31.19.132

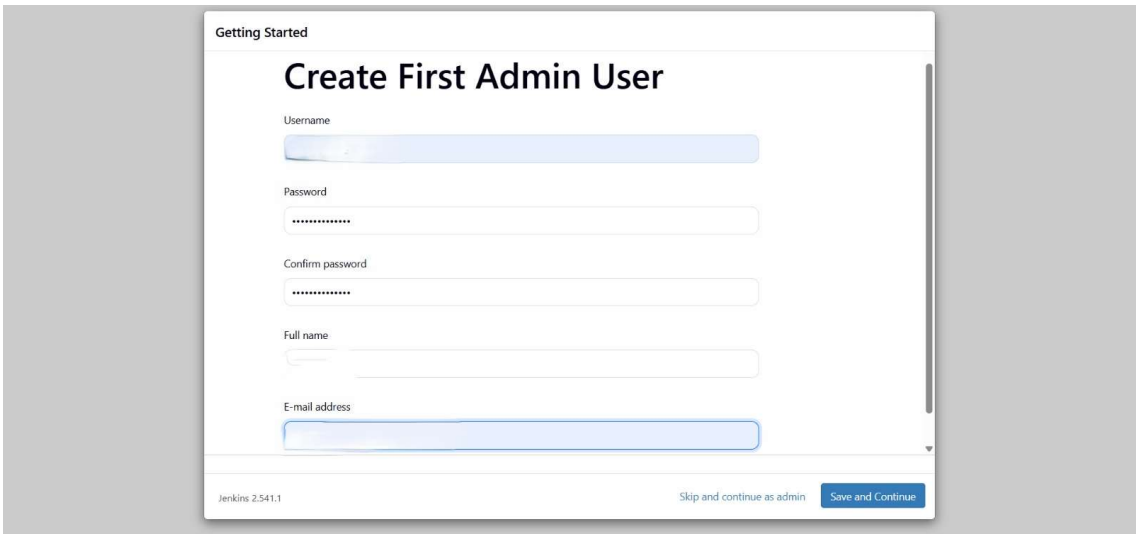
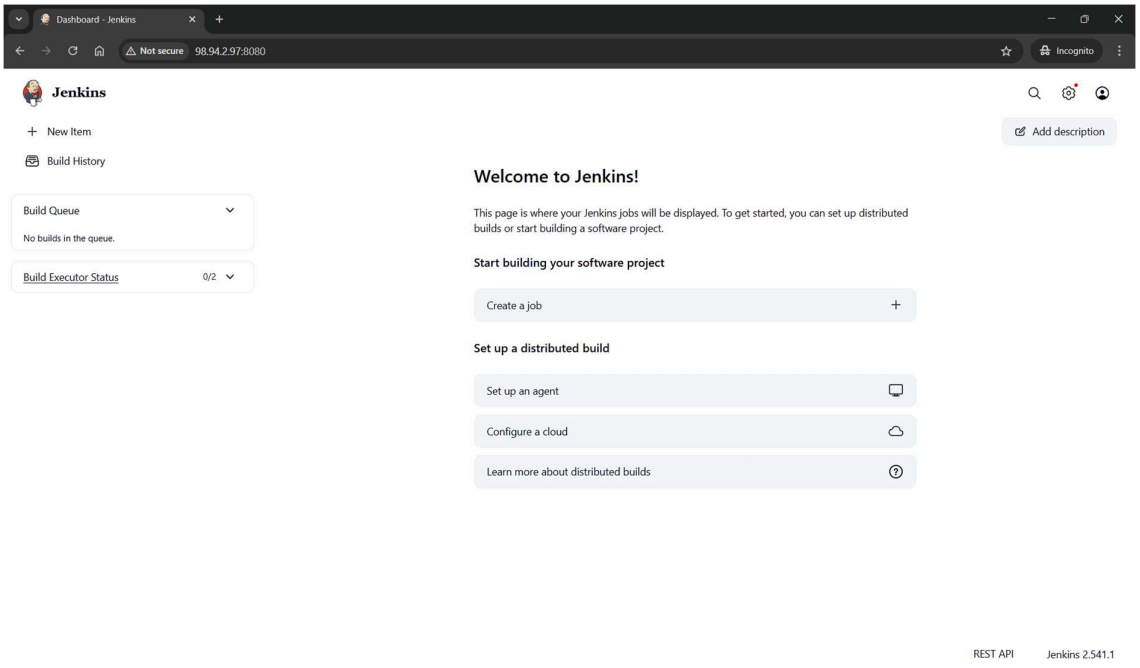
Then we have to install **Jenkins plugins** using the suggested settings.



Then the installation will be done.



Then we have to create an admin user to access the Jenkins.



The Jenkins UI/UX.